ENGINEERING PORTFOLIO

Intelligent Dual-Axis Solar Tracker

Build Progress Supplement — as-built vs proposal, documented corrections, build log & photo record

3	2	8	84
Real LDRs	Servo Axes	Corrections	Build Photos

Anas Mohammed

A2 Level · 2025/2026

CS • PHYSICS • MATHEMATICS

PURPOSE

Build Progress Supplement

This supplement sits alongside the original project proposal (“Intelligent Dual-Axis Solar Tracking System”, Sections 1–7). It does not replace the proposal — the research, evidence base, and problem justification in that document stand. What follows is the as-built reality: what changed between proposal and build, why it changed, and the evidence for it.

A portfolio that only shows the proposal and a finished photo hides the engineering; the value of a DNT/CS portfolio is in showing how the design responded to real constraints. Every deviation below is documented as a deliberate, evidenced decision — not an omission.

9	8	5	84
Components changed	Corrections logged	Build phases	Build photos

01 — AS-BUILT VS PROPOSAL

Specification: Proposal vs. As-Built

Every component or feature that changed between the original evidenced proposal and the final build, with the reason for the change.

Component / Feature	Original Proposal	As Actually Built	Reason for Change
4-LDR sun-tracking pyramid	4x LDR in a pyramid arrangement for full quadrant azimuth/elevation comparison	3x real LDRs (A0=TL, A1=TR, A2=Bottom). Elevation compares avg(TL,TR) vs Bottom instead of true 4-quadrant.	4th LDR never sourced in time; a transistor-based analog mux was designed as a workaround, then dropped as unnecessary once the sensor count was simplified to 3.
INA219 current sensor + live efficiency dashboard	INA219 measures current/voltage; LCD displays live tracked-vs-fixed efficiency gain %	INA219 confirmed dead after full fault-isolation (wiring swap, power test, full I2C scan, line isolation, professional reflow, bare-minimum retest). Replaced with a resistor-divider voltage sensor — voltage only, no current, no power, no efficiency %.	Chip damaged before or during an early solder fault; no replacement sourced — shipping lead time exceeded the deadline window.
GP2Y1010AU0F dust sensor + storm-triggered cover	Dust sensor triggers clamshell cover closure on storm detection; documented hysteresis 0.3–0.5 mg/m3	Dust sensor removed entirely from the final build. Clamshell cover retained, now triggered by night detection (LDR threshold) only.	Sensor produced a persistent flat/zero reading with no response to stimulus after wiring corrections — hardware fault never resolved. Dropped rather than continue debugging dead hardware against the September deadline.
Comidox 6-wire slip ring	Justified as necessary to prevent cable twist during continuous 360° azimuth rotation	Removed. Azimuth is direct-drive, no slip ring.	The justification was factually wrong for this build — the coded azimuth range is 10–170° (a 160° sweep), never continuous rotation. A slip ring solves a problem this design does not have.
Frame material	12mm plywood (main frame) + 3mm acrylic (panel mount, elevation frame)	20mm MDF (enclosure: base, walls, lid, azimuth deck) + 3mm acrylic (elevation frame, panel backing, cover — unchanged)	Switched to MDF for the enclosure/wood components specifically to use reclaimed/discarded offcuts rather than buy new stock. Acrylic components are unchanged from the original plan.
Power source	Single 500mAh 3.7V Li-Po battery via TP4056, powering the whole system	Split-rail architecture: Samsung Li-ion cell (via TP4056 → boost) powers Arduino/LCD/relay/voltage sensor; two separate AA NiMH packs power the four servos in pairs.	Measured, repeatable voltage sag (down to ~3.7V) and Arduino brownout/reset-loop under simultaneous multi-servo load on a single shared rail — confirmed via incremental 1→4 servo load testing, not assumed.
Fixed vs. tracked comparison method	Second identical fixed panel running simultaneously alongside the tracking panel for a direct, controlled comparison	Sequential single-unit test: tracking mode run for a full day, then servos disabled (TRACKING_MODE flag) and the same physical unit run stationary on a separate day; totals compared.	Removes the need for a second INA219/current-sensing circuit (moot now that INA219 is dead anyway). Weather/sun-angle variation between the two test days is an acknowledged confounding variable — see Section 2, item 4.

Component / Feature	Original Proposal	As Actually Built	Reason for Change
Master power switch	Not specified in original proposal	3-pin SPDT slide switch, breadboard-mounted, wired in-line on the TP4056 OUT+ → boost VIN+ path only (OUT – unswitched). Cuts power to Arduino/servos/LCD while leaving the TP4056 free to keep charging the battery.	Added as a safety and usability improvement once the split-rail architecture was finalised — tested and confirmed working.
Servo pin map	Azimuth D9, Elevation D10, Cover D5, Wiper D6	Azimuth D11, Elevation D12, Cover D5, Wiper D6	Azimuth/elevation moved to D11/D12 during the split-rail rebuild; final pin map confirmed and locked.

02 — DOCUMENTED CORRECTIONS

Corrections & Limitations

The items below are proposal-vs-build discrepancies and methodological limitations, stated plainly rather than glossed over. A portfolio that hides these is less credible, not more — examiners and admissions readers can tell the difference between a project that went exactly to plan and one that was actually built.

1. Slip ring justification was factually wrong

The proposal (Section 3.3) justifies the slip ring by claiming it prevents cable twist during continuous 360° rotation. The build's azimuth servo only ever sweeps 10–170° — continuous rotation never occurs. This was identified, and the slip ring removed from the design entirely rather than kept with a retrofitted justification. Direct-drive wiring is used instead, which is also simpler to build.

2. INA219 current sensor is dead; the live efficiency-gain dashboard described in the proposal does not exist in the final build

A full, methodical fault-isolation process (wiring swap, power-delivery LED test, full 1–127 I2C address scan, SDA/SCL line isolation, physical inspection, professional reflow of the suspect joints, and a final bare-minimum 4-wire retest) confirmed the chip itself is internally damaged. No replacement was sourced — international shipping lead times exceeded the remaining build window. The LCD now displays battery voltage and system status only. This is a genuine capability loss from the original proposal, not a cosmetic change, and is stated as such rather than implied to be equivalent.

3. Dust sensor and dust-triggered cover behaviour removed from the final build

The GP2Y1010AU0F was wired per the corrected datasheet pinout, but produced a persistent flat/zero reading with no response to physical stimulus even after the wiring conflict that had been causing false triggers was fixed. With the September deadline approaching, the decision was made to drop dust detection entirely rather than continue debugging unresponsive hardware. The clamshell cover mechanism is retained and fully functional · it now protects against night-time dew/dust settling only, triggered by the existing LDR night-detection threshold, not by an active storm sensor.

4. Fixed-vs-tracked comparison is a sequential single-unit test, not a simultaneous controlled experiment

The tracking panel is run in full tracking mode for one day and totals recorded; servos are then disabled in software (a TRACKING_MODE flag skips trackSun() entirely) and the same physical unit is run stationary on a different day. This is not a controlled experiment — the two conditions never see identical sun angle, cloud cover, or ambient dust at the same moment, which is a genuine confounding variable. This methodology is used because the original plan (a second, simultaneously-running panel with its own INA219) is no longer feasible now that the INA219 is dead. Any efficiency-gain figure produced by this test is presented as illustrative, not as a controlled scientific result.

5. Split-rail power architecture is a deviation from the proposal's single-battery design

The original proposal specifies one 3.7V Li-Po battery powering the whole system via TP4056. Testing showed this could not support the real system: an incremental servo load test (1→4 servos, voltage measured at each step) showed rail voltage collapsing from ~4.8V at rest to as low as 3.7V under 4-servo load, causing Arduino/LCD brownout and reset loops. The system now runs on three independent supplies — a Samsung Li-ion cell (via TP4056 and a boost converter) dedicated to logic (Arduino, LCD, relay, voltage sensor), and two separate AA NiMH battery packs dedicated to servo pairs. This is a measured, evidence-driven engineering trade-off, not an oversight: it adds complexity and a second/third power source, but it was necessary to get a stable running system.

6. Enclosure material changed from plywood to MDF

The original proposal specifies 12mm plywood for the main frame. The build enclosure (base, walls, lid, azimuth deck) uses 20mm MDF instead, built from reclaimed/discarded offcuts rather than newly purchased stock — a deliberate resourcefulness and cost decision. This carries a real engineering trade-off worth stating honestly: MDF absorbs moisture and is more prone to swelling/delamination than plywood, which matters for an enclosure that may sit outdoors. The acrylic components (elevation frame, panel backing, cover flaps) are unchanged from the original proposal.

OUTSTANDING: whether the MDF surfaces have been sealed (varnish, paint, or similar) against moisture is not yet confirmed. Seal before final submission, or state plainly here that sealing is still planned — do not submit implying it is done if it isn't.

7. TP4056 charging module failure and replacement

During live wiring, a loose battery-negative connection caused a spark. Power was disconnected immediately; the module was inspected for physical damage and, despite no visible burn marks, an unusual electronic/burnt smell was detected. As a precaution the module was retired rather than reused. A replacement TP4056 was sourced, resoldered (B+/B–/OUT+/OUT–), and reconnected into the charging chain. This is included as a genuine safety incident and a demonstration of appropriate caution around lithium battery charging circuits, not omitted.

8. 3-LDR tracking is a simplification from the proposed 4-LDR pyramid

Only 2 of the originally planned 4 LDR positions were ever populated with real sensors; a third was added later. Elevation tracking now compares the average of the two top sensors (TL, TR) against a single bottom sensor, rather than a genuine 4-quadrant comparison. This trades some tracking precision for reliability and buildability within the project timeline. An earlier attempt to work around the missing sensor via a 5-transistor discrete multiplexer was designed, coded, and then abandoned as unnecessary once the sensor count was simplified to 3 and the pin budget resolved itself.

03 — BUILD LOG

Progress Journal

A condensed, phase-organised account of the build, drawn from the full session-by-session fault-diagnosis and testing history.

PHASE A — DRY BUILD & COMPONENT BRING-UP

Phase A — COMPLETE

Arduino, LCD, and initial sensor bring-up

Confirmed Arduino UNO (CH340 clone) communicating correctly with the IDE after resolving a driver issue. LCD confirmed as I2C type, found at address 0x27, contrast fixed via the backpack potentiometer. First two LDRs wired and tested on A0/A1 with 4.7kohm resistors — confirmed a ~300 ADC-count swing between lit and covered states, validating the tracking algorithm's 50-count tolerance threshold. All four relay channels confirmed active-LOW. All four servos individually tested on their original pins with clean 0–180° sweep.

Phase A — COMPLETE

Soldering at school

INA219 header pins, TP4056 wire leads (input, B+/B–), and Samsung battery leads were all soldered at school, making all three components breadboard-ready. The dust sensor's 150ohm resistor and 220uF capacitor were sourced separately after discovering they were not included in the sensor packaging as the manufacturer documentation claimed.

PHASE B — FAULT DIAGNOSIS: INA219, DUST SENSOR, MUX REDESIGN

Phase B — COMPLETE

Boost calibration and first USB charging success

The MT3608 boost converter was calibrated to 5.01V using a multimeter and successfully powered a phone to a confirmed charging state via the USB output module — the first full charging-chain validation.

Phase B — ISSUE

INA219 diagnosed as dead

A structured fault-isolation process — wiring swap, power-delivery LED test, full I2C address scan (1–127, nothing found at any address while the LCD remained reliably found at 0x27), SDA/SCL line isolation (revealing an asymmetric fault pointing at the SDA line), physical inspection of the solder joints, professional reflow, and a final bare-minimum 4-wire retest — concluded the chip itself is internally damaged. Replaced with a resistor-divider voltage-only sensor.

Phase B — ISSUE

Dust sensor wiring corrected, then found unresponsive

A wiring guide audit caught the dust sensor's V-LED, LED-control, and S-GND lines wired to the wrong pins entirely (would have put 5V on the sensor's ground reference). Corrected per datasheet. Even after correction, the sensor produced a flat, unresponsive reading, pointing to a hardware fault under active investigation at this stage.

Phase B

COMPLETE

Transistor multiplexer designed, then abandoned

With all 6 analog pins already committed, a discrete 4-channel multiplexer was designed from 5 BC547 transistors to free up pins for the remaining LDRs. It was built and coded, then made unnecessary once the sensor count was simplified to 3 real LDRs and the dust sensor was dropped — removing the pin-count pressure the mux was built to solve.

PHASE C — SIMPLIFICATION & PHYSICAL SOLDERING

Phase C

COMPLETE

Dust sensor and multiplexer dropped from the design

Given the dust sensor's unresolved hardware fault and the approaching deadline, dust detection was dropped entirely rather than continuing to debug dead hardware. The clamshell cover and dawn wiper were both retained, now triggered by night detection only. All four solar panels and three LDRs were physically soldered this phase, surfacing the importance of proper strain relief before frame assembly.

Phase C

ISSUE

Power source evaluation: Samsung cell rejected, 18500 cell adopted then also rejected

A salvaged 18500 Li-ion cell (900mAh) was adopted over a phone battery, which was rejected on safety grounds (bare pouch-cell tabs, no visible protection circuit). The 18500 cell later failed to respond during testing; root cause was narrowed to either a genuinely failed cell or a latched protection-circuit cutoff, not resolved with available equipment at the time. The phone battery (Samsung EB-BG360CBN) was reinstated as the interim working power source, and the full charging chain — battery, TP4056, boost converter, USB module, phone — was re-verified successfully.

PHASE D — POWER SYSTEM DEBUGGING & SPLIT-RAIL REBUILD

Phase D

COMPLETE

Master switch fault found and replaced

A planned master switch was diagnosed — through a structured multimeter test including a control check — as having failed internal contacts, and was rejected as unfit for use as a safety-critical cutoff on an unprotected lithium battery. A replacement was later sourced and tested.

Phase D

COMPLETE

Servo brownout root-caused to shared-rail current capacity, not wiring

An incremental load test (1→4 servos, voltage measured at each step) showed rail voltage collapsing from ~4.8V at rest to ~3.7V under full load — ruling out breadboard contact faults and individual bad servos in favour of a genuine current-capacity limitation in the single boost/battery supply. Firmware mitigations (sequencing, settle delays) reduced but did not eliminate the sag, confirming the fix had to be architectural.

Phase D

COMPLETE

Split-rail architecture implemented

The system was rebuilt onto three independent supplies: the Samsung Li-ion cell (via TP4056 and boost) for Arduino/LCD/relay/voltage sensor, and two separate battery packs for the four servos in pairs. Zinc-carbon AA cells were tested and confirmed unsuitable (high internal resistance causing voltage collapse under load) and replaced with NiMH cells, which showed an immediate, measurable improvement under the same load conditions. A swapped power/ground connector on one MG90 servo — found to be simultaneously causing servo heat and LCD garbling across several earlier sessions — was identified and fixed.

PHASE E — TRACKING FIX, MASTER SWITCH, MDF ENCLOSURE BUILD

Phase E

COMPLETE

Dual-axis tracking bug root-caused and fixed

A persistent elevation-tracking fault, investigated across multiple sessions under a wiring/pin hypothesis, was found to actually be a software interlock (an azimuthMoved boolean) blocking elevation correction whenever azimuth had already moved in the same loop pass. Fixed by letting both axes correct independently each pass. Azimuth and elevation servos were moved to their final pins, D11 and D12, and both confirmed physically moving.

Phase E

COMPLETE

Master switch installed and tested

The replacement 3-pin SPDT switch was wired in-line on the TP4056 OUT+ → boost VIN+ path (OUT – left unswitched, so the battery keeps charging regardless of switch state), mounted on the breadboard rather than soldered for continued flexibility during build. Tested and confirmed working.

Phase E

IN PROGRESS

Enclosure switched to MDF; azimuth deck enlarged

The enclosure was redesigned around 20mm MDF (using reclaimed material) at 200×200×100mm external dimensions, with the azimuth deck enlarged from the original 120mm to a 140mm corner-to-corner octagon to better fit the larger enclosure. Acrylic components (elevation frame, panel backing, cover) are unchanged. Base-level fabrication is complete as of this entry; remaining walls, lid, and mechanical assembly are in progress.

03 — REFERENCE

Final Servo / Sensor Pin Map

Component	Pin	Function
Azimuth MG90S	D11	Rotates upper assembly
Elevation MG90S	D12	Tilts panel frame
Cover SG90	D5	Clamshell cover
Wiper SG90	D6	Dawn sweep
LDR TL	A0	Light sensing
LDR TR	A1	Light sensing
LDR Bottom	A2	Light sensing
Voltage sensor	A3	Battery voltage
LCD SDA	A4	I2C data
LCD SCL	A5	I2C clock
Relay CH1	D4	USB charge control
Relay CH2–4	D2/D3/spare	Unused

MDF CUTTING LIST (20mm MDF)

Part	Qty	Size (mm)	Notes
Base floor	1	200 × 200 mm	Main enclosure base
Long walls	2	200 × 100 mm	20 mm MDF stock
Short walls	2	160 × 100 mm	Between long walls
Fixed lid	1	200 × 200 mm	Top/lid
Battery-bay lid	1	≈70 × 70 mm	Removable / screwed
Azimuth deck	1	140 mm octagon	Corner-to-corner; side ≈53.6 mm
Azimuth servo block	1	≈30 × 25 × 20 mm	D11 support

Enclosure target: 200 × 200 × 100 mm external. Azimuth octagon: 140 mm corner-to-corner, radius 70 mm, side ≈53.6 mm. Acrylic components (elevation arms/base/tab, panel backing, LDR mounts, cover flaps and hinge reinforcement) are unchanged from the original proposal.

APPENDIX

Build Photo Log

All 84 build photos, captioned and grouped by build phase to match the log above.

PHASE A — DRY BUILD & COMPONENT BRING-UP

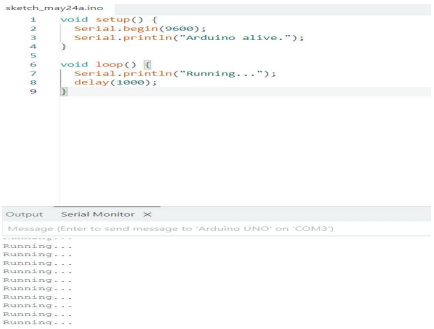


Fig. 1
Arduino Serial Monitor: “sketch_may24a.ino” — setup()/loop() printing “Arduino alive.” / “Running...” — first Arduino comms confirmation



Fig. 2
Arduino UNO board, close-up, powered (green LED lit)

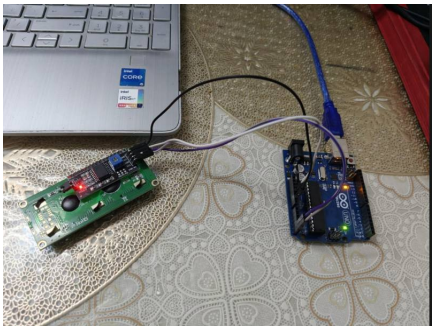


Fig. 3
LCD + Arduino wired together on jumpers, LCD showing red backlight only (early bring-up, pre-fix)

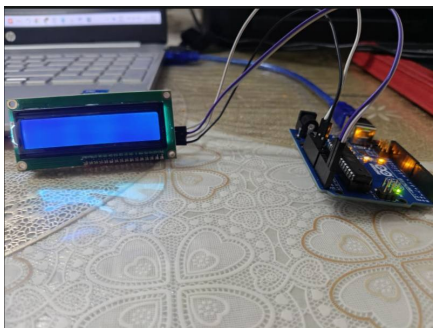


Fig. 4
LCD showing blue backlight, no text visible — backlight-only fault stage



Fig. 5
I2C scanner sketch + Serial output: “Device found at: 0x27” — confirms LCD I2C address

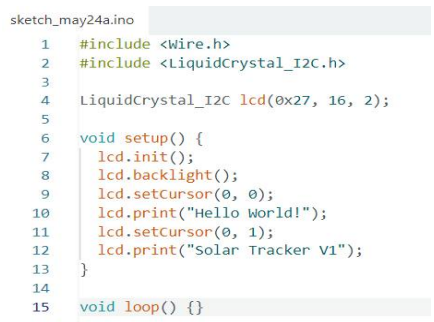


Fig. 6
LiquidCrystal_I2C test sketch: lcd(0x27,16,2), prints “Hello World!” / “Solar Tracker V1”

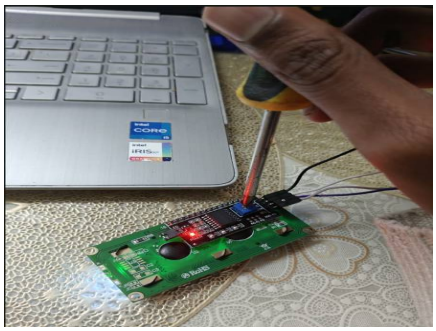


Fig. 7
Adjusting LCD backpack contrast potentiometer with a screwdriver (fixing the blank-text issue)



Fig. 8
LCD now correctly displaying “Hello World! / Solar Tracker V1” — fix confirmed

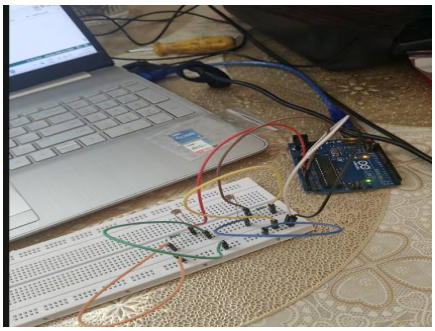


Fig. 9
4-channel relay module wired to breadboard + Arduino for relay bring-up test

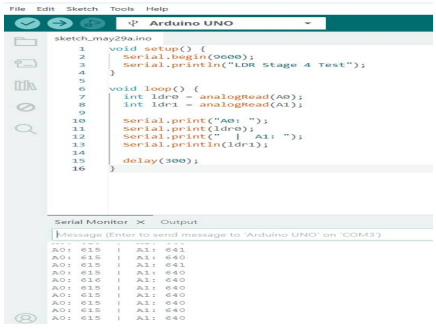


Fig. 10
Serial output “LDR Stage 4 Test” sketch — A0:615 | A1:640ish baseline readings

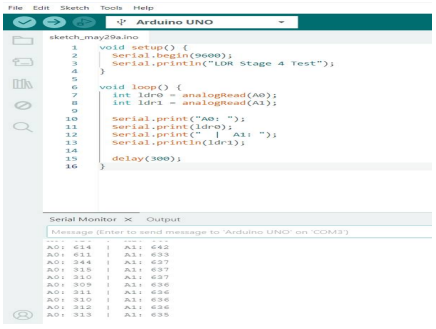


Fig. 11
Same sketch, LDR A0 covered — reading drops to ~310 (confirms the ~300-count swing used to set LDR_TOLERANCE=50)

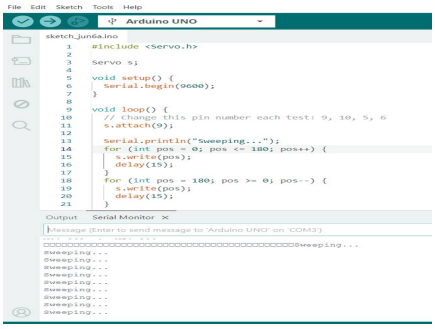


Fig. 12
Servo sweep test sketch — comment “Change this pin number each test: 9,10,5,6”, sweeping 0-180-0

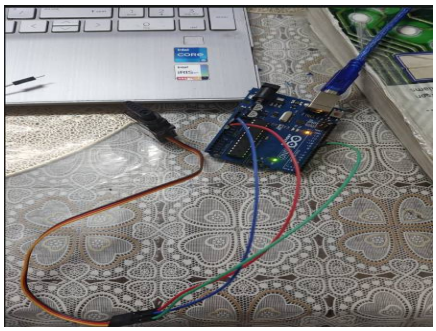


Fig. 13
Single servo wired directly to Arduino UNO for the sweep test

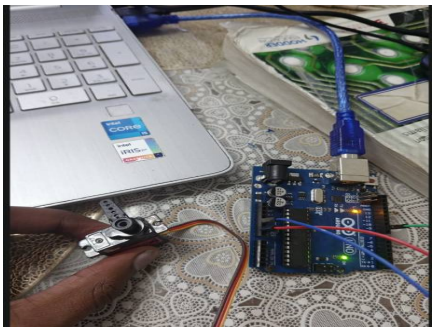


Fig. 14
Hand holding an MG90S mid-sweep

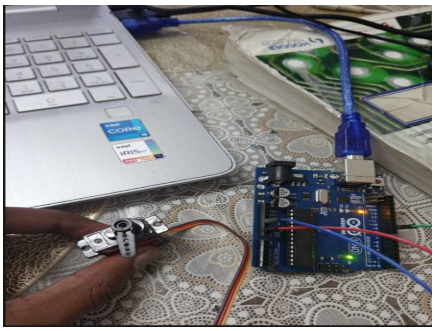


Fig. 15
Same servo test, different angle

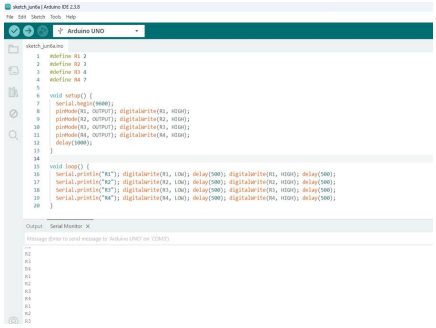


Fig. 16
Relay cycle test sketch: R1-R4 defined as pins 2,3,4,7, cycling each with Serial.println

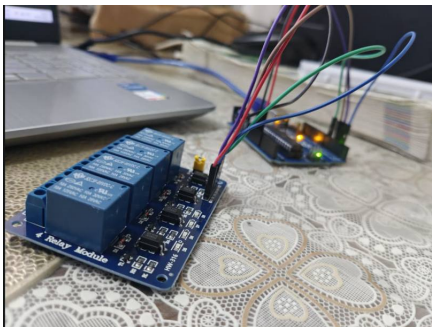


Fig. 17
4-relay module fully wired to Arduino, all 4 channels connected, LEDs lit

PHASE B — FAULT DIAGNOSIS: INA219, DUST SENSOR, MUX REDESIGN

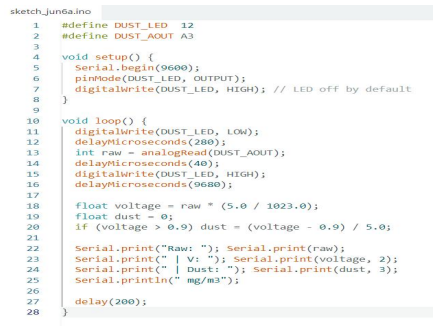


Fig. 18
Dust sensor test sketch: DUST_LED=12, DUST_AOUT=A3, 280us LED pulse timing, mg/m3 calc

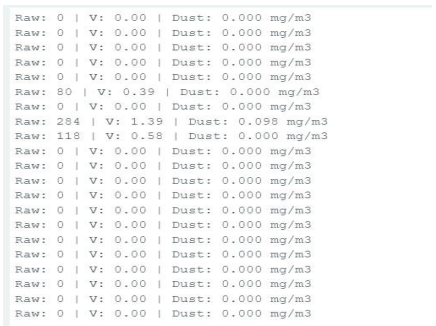


Fig. 19
Dust sensor Serial output — baseline clean-air readings, mostly “Raw: 0 | Dust: 0.000 mg/m3”

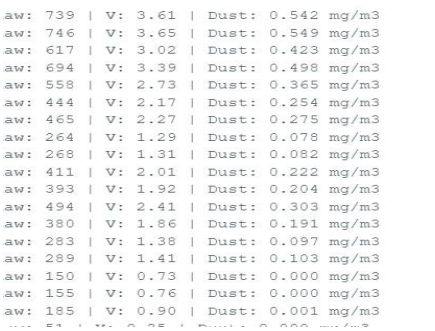


Fig. 20
Dust sensor Serial output — blow test peak “Raw: 746 | V: 3.65 | Dust: 0.549 mg/m3” (exceeds 0.5 storm threshold)

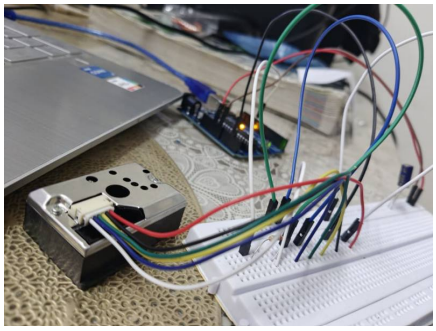


Fig. 21
GP2Y1010AU0F dust sensor unit wired into breadboard, laptop in background

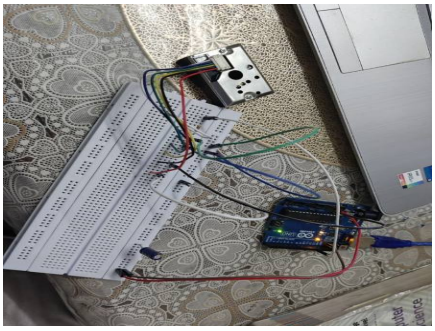


Fig. 22
Full dust sensor circuit on breadboard, electrolytic capacitor visible

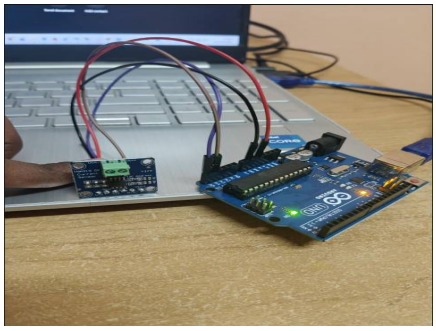


Fig. 23
INA219 current sensor being wired to Arduino (header pins already soldered)

```
Voltage: 1.03 V  |  Current: -0.20 mA
Voltage: 1.03 V  |  Current: -0.20 mA
Voltage: 1.03 V  |  Current: -0.40 mA
Voltage: 1.03 V  |  Current: -0.20 mA
Voltage: 1.03 V  |  Current: -0.30 mA
Voltage: 1.03 V  |  Current: -0.20 mA
Voltage: 1.03 V  |  Current: 0.10 mA
Voltage: 1.03 V  |  Current: -0.40 mA
Voltage: 1.03 V  |  Current: -0.10 mA
Voltage: 1.03 V  |  Current: -0.30 mA
```

Fig. 24
INA219 early Serial output: “Voltage: 1.03 V | Current: -0.20 mA” (noisy near-zero current, flat voltage)

sketch_jun16a.ino

```ino
#include <Wire.h>
#include <Adafruit_INA219.h>

Adafruit_INA219 ina219;

void setup() {
  Serial.begin(9600);
  Wire.begin();
  if (!ina219.begin()) {
    Serial.println("INA219 not found. Check wiring.");
    while (1);
  }
  Serial.println("INA219 found.");
}

void loop() {
  float voltage = ina219.getBusVoltage_V();
  float current = ina219.getCurrent_mA();
  Serial.print("Voltage: "); Serial.print(voltage); Serial.print(" V  |  ");
  Serial.print("Current: "); Serial.print(current); Serial.println(" mA");
  delay(1000);
}
```

Fig. 25
INA219 test sketch using Adafruit_INA219 library, “INA219 not found. Check wiring.” handler

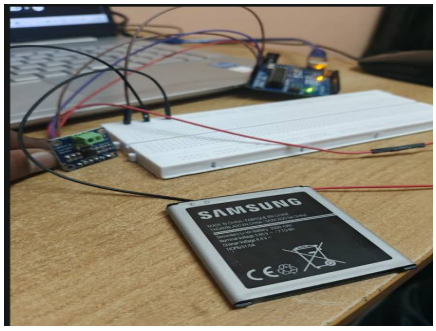


Fig. 26
Samsung EB-BG360CBN battery wired to INA219+breadboard — label visible: 2000mAh, 4.4V charge, 3.85V nominal

```
Bus Voltage: 3.71 V
Bus Voltage: 3.71 V
Bus Voltage: 3.70 V
Bus Voltage: 3.71 V
Bus Voltage: 3.71 V
Bus Voltage: 3.71 V
Bus Voltage: 3.71 V
```

Fig. 27
Voltage sensor Serial output: “Bus Voltage: 3.71 V” readings on the Samsung battery

```ino
#include <Wire.h>
#include <Adafruit_INA219.h>

Adafruit_INA219 ina219;

void setup() {
  Serial.begin(9600);

  if (!ina219.begin()) {
    Serial.println("INA219 not found");
    while (1);
  }

  Serial.println("INA219 ready");
}

void loop() {
  Serial.print("Bus Voltage: ");
  Serial.print(ina219.getBusVoltage_V());
  Serial.println(" V");

  delay(1000);
}
```

Fig. 28
Revised INA219 test sketch — voltage-only print loop, “INA219 ready” handler

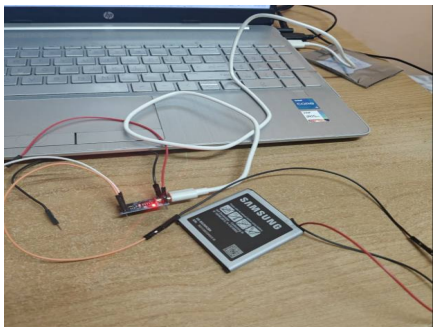


Fig. 29
TP4056 module (red charging LED lit) wired to Samsung battery — later session, different desk

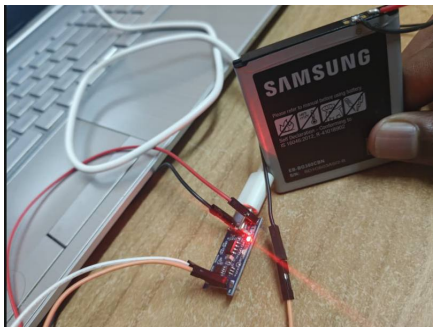


Fig. 30
Same TP4056 test, closer angle — Samsung battery model label “EB-BG360CBN” clearly visible

```
Battery Voltage: 3.74 V
Battery Voltage: 3.74 V
Battery Voltage: 3.75 V
Battery Voltage: 3.74 V
Battery Voltage: 3.75 V
Battery Voltage: 3.75 V
Battery Voltage: 3.75 V
Battery Voltage: 3.75 V
```

Fig. 31
Serial output “Battery Voltage: 3.74-3.75 V” — resistor-divider voltage sensor readout

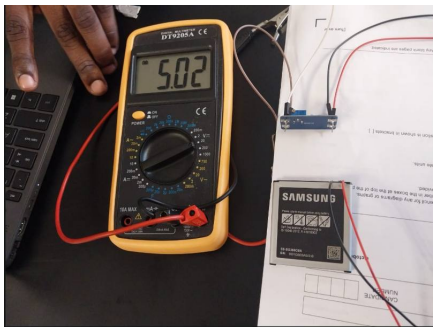


Fig. 32
DT9205A digital multimeter reading 5.02V — boost converter calibration confirmation

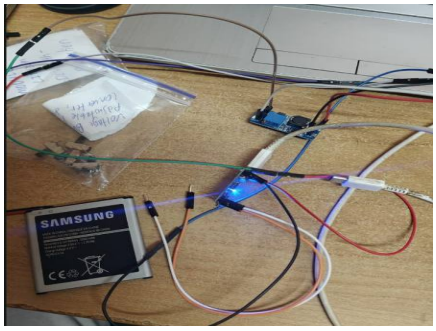


Fig. 33
USB output module with blue LED lit (confirmed active 5V output) + labeled “Voltage Boost Converter” parts bag

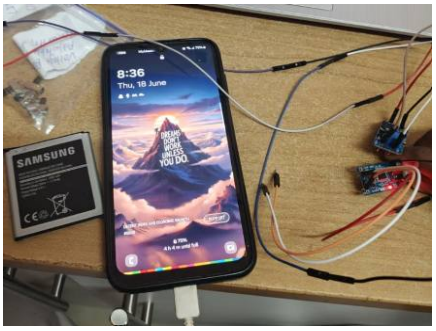


Fig. 34
Phone lock screen, “8:36, Thu 18 June” — charging session date confirmation



Fig. 35
Phone showing “75% — 4h 5m until full” while wired to USB output module

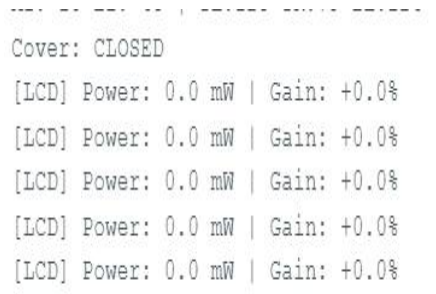


Fig. 36
Serial output from an early full-system build: “Cover: CLOSED” / “[LCD] Power: 0.0 mW | Gain: +0.0%” — original INA219 efficiency-dashboard code still active, pre-dropout

PHASE C — SIMPLIFICATION & PHYSICAL SOLDERING

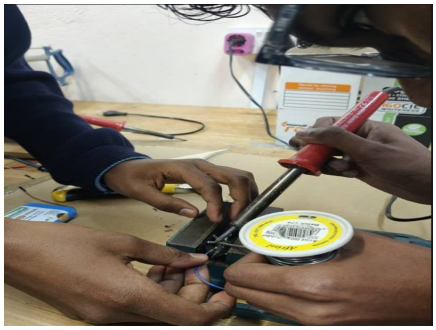


Fig. 37
Soldering session at school (DNT lab), assisted, close-up of iron on a wire



Fig. 38
Same school soldering session, different framing, glasses/safety visible

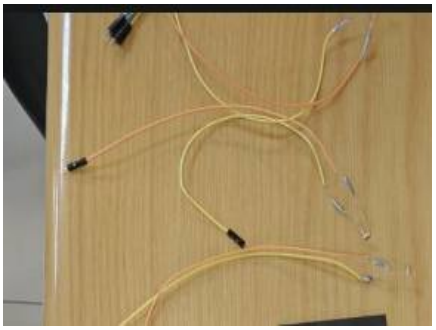


Fig. 39
Freshly soldered wire leads with Dupont connectors on a wooden desk

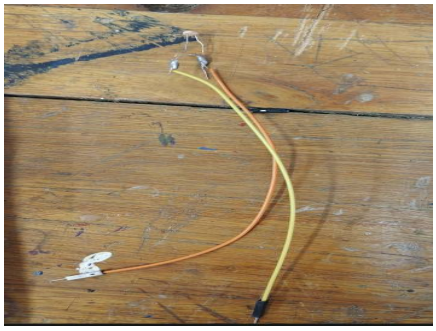


Fig. 40
Two soldered leads (orange/yellow) with heat-shrink/connector ends



Fig. 41
Soldering the Samsung battery's connector wires — flat pouch cell visible

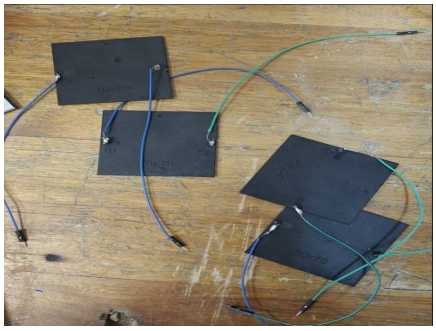


Fig. 42
All four 70x70mm solar panels laid out with soldered output leads (blue/green wire pairs)



Fig. 43
Close-up: soldering a diode/lead onto a panel's positive terminal

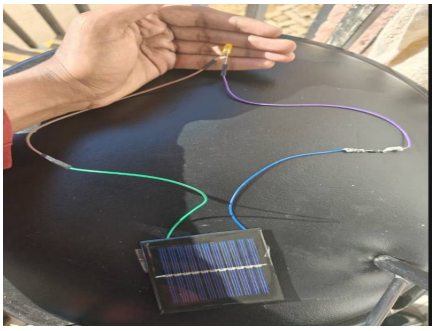


Fig. 44
Outdoor LED continuity test, panel #1, wires held up (no LED lit yet in this frame)

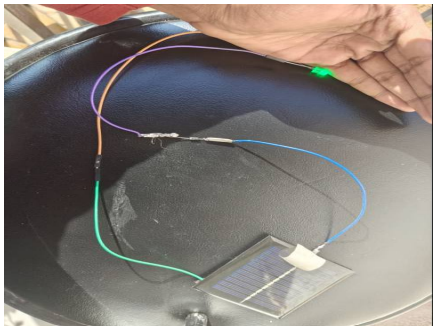


Fig. 45
Outdoor LED test, panel visible — GREEN LED lit, confirms power in direct sun

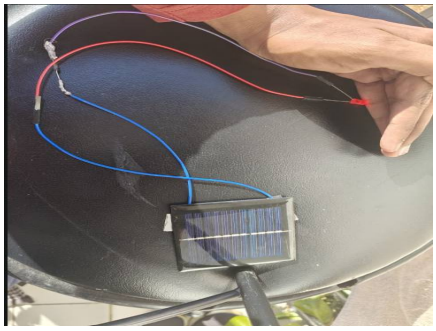


Fig. 46
Outdoor LED test, second panel — RED LED lit

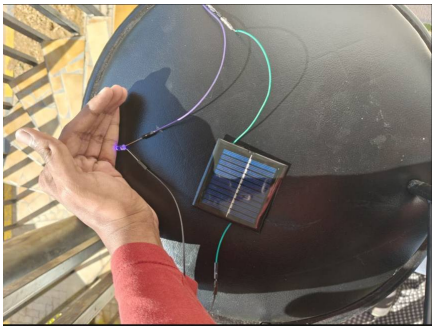


Fig. 47
Outdoor LED test, third panel, balcony railing — PURPLE/VIOLET LED lit



Fig. 48
The salvaged 18500 Li-ion cell, label clearly reads “18500/900mAh 3.7V”



Fig. 49
The toy RC vehicle the 18500 cell was salvaged from, battery still in its compartment

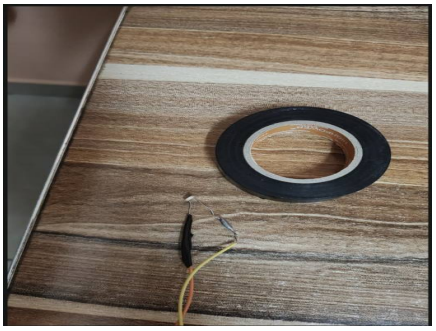


Fig. 50
Roll of PVC electrical tape + a partially-insulated solder joint on a wood-grain desk

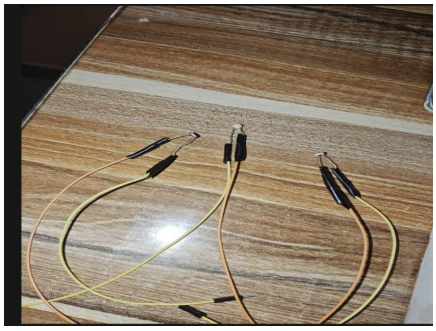


Fig. 51
Three soldered/heat-shrunk wire joints laid out, close-up



Fig. 52
Soldered/tinned joint between a crocodile-clip lead and a jumper wire



Fig. 53
All four solar panels laid out, output leads wired and joints wrapped in black PVC tape

PHASE D — POWER SYSTEM DEBUGGING & SPLIT-RAIL REBUILD



Fig. 54
4xAA holder (zinc-carbon cells) wired to breadboard with slide switch, resistor and LED — switch off

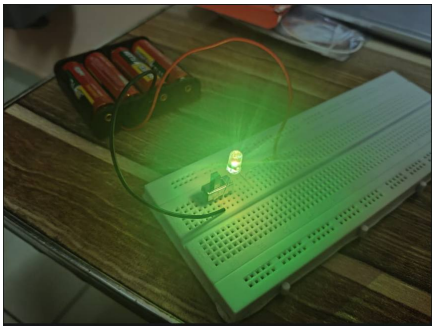


Fig. 55
Same rig, switch on — LED lit



Fig. 56
Same rig, second frame, LED lit (blue cast) — switch test repeated



Fig. 57
Multimeter with the 3-pin slide switch held on the dial — switch continuity check before installing

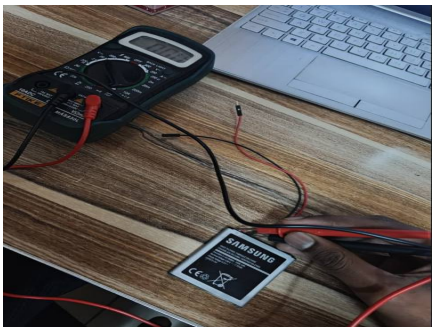


Fig. 58
Multimeter probes on the Samsung battery terminals, laptop in background



Fig. 59
Multimeter reading 3.72 V on the Samsung battery

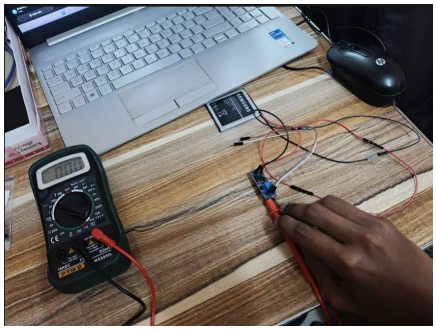


Fig. 60
Samsung battery wired to the boost-converter board, multimeter reading 0.00 V at that moment

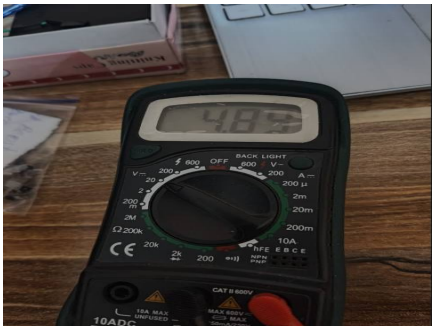


Fig. 61
Multimeter reading 4.89 V — boost converter output after bench test

PHASE E — TRACKING FIX, MASTER SWITCH, MDF ENCLOSURE BUILD



Fig. 62
MDF sheet with printed final build blueprint and cutting list page, tape measure — start of fabrication



Fig. 63
Marking cut lines on MDF with steel square and tape measure, blueprint alongside, panel saw behind



Fig. 64
MDF board marked with part labels and cut lines, positioned on the table saw



Fig. 65
Ripping a marked MDF strip on the table saw



Fig. 66
Cutting a marked MDF panel on the table saw

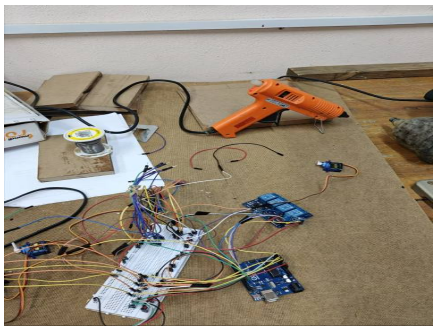


Fig. 67
Workshop bench with full breadboard prototype (Arduino, relay module, servos) beside a hot glue gun

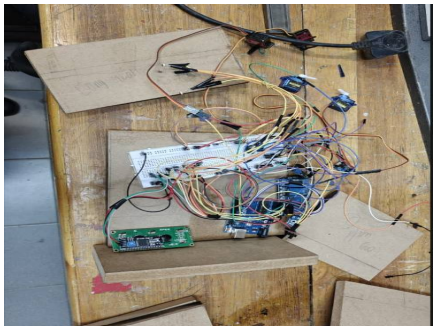


Fig. 68
Same prototype on a workshop bench, LCD placed on an MDF panel, part templates around it

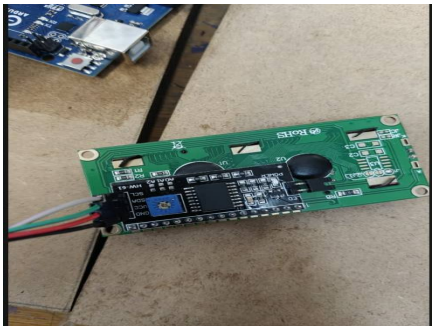


Fig. 69
Rear of the 16x2 LCD: I2C backpack (HW-61) with SCL/SDA/VCC/GND wires and contrast trimmer



Fig. 70
Drilling an MDF panel on the drill press



Fig. 71
Cutting an MDF panel with a jigsaw, workpiece clamped

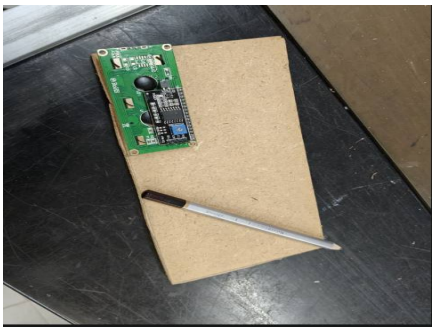


Fig. 72
LCD held against a marked MDF panel to size the window cutout

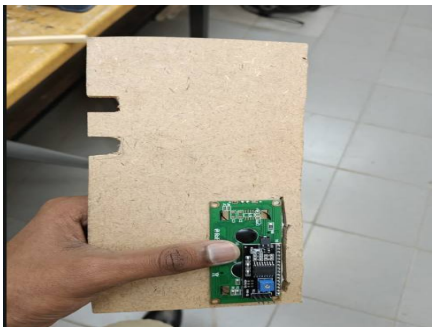


Fig. 73
MDF LCD panel with notches cut and LCD placed on it



Fig. 74
Same panel beside a cordless drill, LCD in place



Fig. 75
Cutting an MDF piece on a scroll saw



Fig. 76
Square MDF panel marked with diagonals, blueprint and steel ruler on bench, a servo being measured

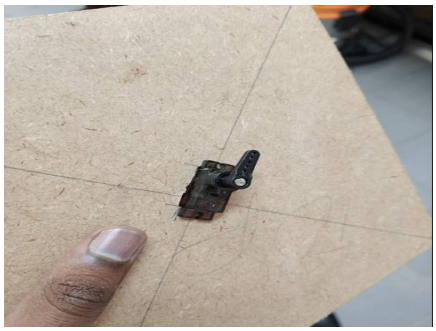


Fig. 77
Servo placed at the diagonals' intersection on the MDF panel to mark its mounting position

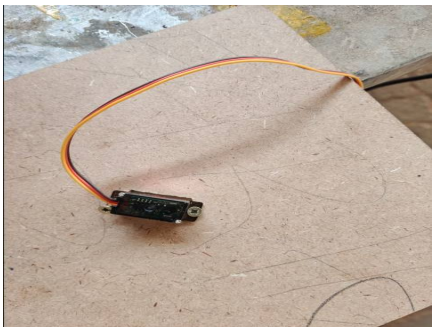


Fig. 78
Servo sitting on the panel, wire looped



Fig. 79
Jigsaw cutting a square opening in an MDF piece, four corner holes pre-drilled, clamped

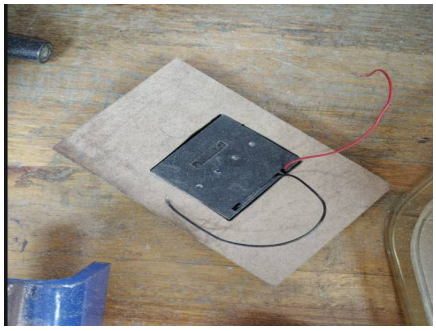


Fig. 80
4xAA battery holder placed on an MDF piece to plan its mounting position



Fig. 81
MDF panel with a cut in progress (clamped), cardboard octagon template alongside



Fig. 82
Cardboard octagon template with radial layout lines, clamped, next to a cordless drill



Fig. 83
Working a marked octagonal MDF piece with a hammer at the school workshop

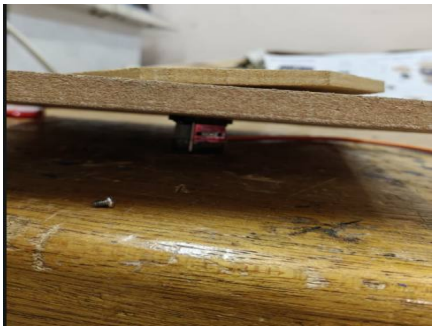


Fig. 84
Servo mounted underneath a layered MDF deck, loose screw on the bench